

Transfer Learning for Improving Neuroimaging-Based Diagnostic Classification

Gopikrishna Deshpande , Senior Member, IEEE, Bonian Lu, Nguyen Huynh , and D Rangaprakash

Abstract—Overfitting, an issue that constrains the validity and generalizability of machine learning in neuroimaging-based diagnostic-classification, is in part due to small sample sizes in relation to what is required for generalization. Even with data aggregation (such as in Autism Brain Imaging Data Exchange or ABIDE), the relatively smaller sample sizes are a result of the fact that it is difficult/expensive to acquire data from clinical populations. With healthy controls, we have comparatively larger samples available. Therefore, we propose to address overfitting by using larger healthy samples (from Human Connectome Project or HCP) to learn the diversity of neural signatures of healthy controls, with the aim of transferring that learning into the context of discriminating autism from healthy controls in ABIDE. **Methods:** We developed a complete variational autoencoder based transfer learning framework including data oversampling, model pre-training, classifier training and testing, and model explanation. Then, the performance of transfer learning was estimated and visualized. **Results:** The transfer learning classification model achieved about 7% more accuracy on site-mismatched data than obtained without transfer learning. **Discussion:** Overall, we demonstrated the applicability of transfer learning within a deep learning framework for utilizing larger samples of available healthy control data to improve generalizability and accuracy of diagnostic classification in ASD, as well as reduce the harmful effects of inter-site variability on classification. We believe the proposed framework is potentially applicable to other disorders as well.

Index Terms—Autism spectrum disorders, functional connectivity, transfer learning, variational autoencoder, layer-Wise relevance propagation.

I. INTRODUCTION

THE use of artificial intelligence in diagnosing psychiatric disorders based on neuroimaging data is an attractive proposition as it provides an element of objectivity in otherwise traditionally subjective mode of diagnosis. Despite a strong logical case, its practical implementation and utility remains a challenge due to limited generalizability and prediction accuracy. Here we focus on one such mental disorder, autism spectrum disorder (ASD), which is clinically diagnosed by impairment in communications and social skills [1], [2], [3], [4]. Deep learning (DL) is a recent data driven approach to build hierarchical multi-layer structures to automatically predict outputs of a given system from novel inputs [5], [6].

Deep learning models have been increasingly applied in neuroimaging, particularly magnetic resonance imaging (MRI), to improve the diagnostic accuracy of brain disorders. For example, Al-Adhaileh et al. [7] used high-resolution MRI data (or anatomical scans) in combination with convolutional neural networks (CNNs) to classify Alzheimer's disease (AD) and achieved good accuracy. Li et al. [8] proposed a graph neural network model to capture the complex structure of functional brain networks. Their model extracts graph representation features and utilizes a clustering method to identify key brain regions with potential as biomarkers for brain disorders. Additionally, to address the issues of data scarcity, several generative deep learning models have been proposed in the neuroimaging field. A generative adversarial network that learns the distribution of functional MRI data on ADHD was proposed by Qiang et al. [9] to synthesize data and create a balanced dataset. They also controlled the instability of the GAN by incorporating it into the autoencoder framework. Among the various deep learning architectures, Transformer-based models have emerged as powerful tools for capturing complex relationships within data using self-attention mechanisms. Their ability to learn effective feature representations from large-scale datasets has led to their adoption in various brain research and diagnostic applications [10], [11]. However, like GCN or CNN, while Transformers demonstrated promising performance within the dataset from a single site, their generalizability to datasets from other sites, beyond the training site, remains limited. Additionally, their interpretability in network neuroscience is a significant concern, as post-hoc

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Gopikrishna Deshpande is with the Auburn University Neuroimaging Center, Department of Electrical and Computer Engineering, Auburn University, Auburn, AL 36849 USA, also with the Department of Psychological Sciences, Auburn University, Auburn, AL 36849 USA, also with the Alabama Advanced Imaging Consortium, Birmingham, AL 35294 USA, also with the Center for Neuroscience, Auburn University, AL 36849 USA, also with the Department of Psychiatry, National Institute of Mental Health and Neurosciences, Bangalore 560029, India, and also with the Department of Heritage Science and Technology, Indian Institute of Technology, Hyderabad 502284, India (e-mail: gopi@auburn.edu).

Bonian Lu and Nguyen Huynh are with the Auburn University Neuroimaging Center, Department of Electrical and Computer Engineering, Auburn University, Auburn, AL 36849 USA.

D Rangaprakash is with the Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital, Harvard Medical School, Boston, MA 02129 USA.

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interpretability strategies often yield inconsistent and unreliable outputs, making it challenging to derive meaningful insights from model decisions. Therefore, in this study, we proposed a transfer learning framework designed to transform knowledge learned from one site and apply it to others, along with a robust interpretability method to enhance the reliability of model explanations.

A. Limitations of Existing Deep Learning Approaches

Despite the varying degrees of success of DL techniques in fields of computer vision, speech recognition, and natural language processing [12], [13], [14], its applicability in the context of neuroimaging based diagnosis of mental disorders is yet unclear, primarily due to questions about overfitting and generalizability. For example, using open-source large data such as Autism Brain Imaging Data Exchange (ABIDE) [15], several papers found DL techniques to outperform traditional machine learning methods in ASD classification research [16], [17]. However, overfitting is arguably the first issue that constraints the performance of deep neural networks in this field [18], [19]. One major cause for overfitting is a combination of small sample size and relatively high dimensionality in the training dataset, which is also known as the curse-of dimensionality or small-n-large-p problem [20].

Another issue limiting the performance of deep neural networks (DNN) is the heterogeneity between training and test dataset, especially when datasets are collected from different sites as in large open-source databases. Even though previous studies have used relatively large training samples, the prediction accuracy from multisite data was significantly lower than that from single site data [19], [21], [22]. For large open-source data such as ABIDE I, the data are pooled from 18 acquisition sites with different scanner types, acquisition parameters, age/gender distribution of subjects [23]; these lead to different data distributions from different sites. However, in real clinical circumstances, test data will probably be acquired from different institutions than the ones on which the classifier is trained. A classifier cannot easily predict novel data distributions based on different data distributions in the training dataset. A prior study using ABIDE data has reported that classification accuracy can be significantly degraded when training and testing data are drawn from different sites as compared to them being drawn equally from those sites [19]. Thus, prior research indicates that the multi-site data requires DL classifiers to be able to generalize better [22], [24].

B. Transfer Learning in Neuroimaging

A straightforward solution to reduce overfitting is combining the knowledge learned from datasets in other source domains and fine-tuning an adaptive neural network by training fewer examples from the current target domain. This principle of “transferring” knowledge learned from one dataset into another is known as transfer learning (TL) and is a dominant approach in multiple fields of science and engineering [25], [26], and has the potential to be applied in neuroimaging-based diagnostics. Previous neuroimaging studies have used TL in different contexts [27], [28]. One study [29] applied a novel two-path

3D convolutional neural network (CNN) architecture, to train structural and functional MRI simultaneously. They achieved an accuracy of 69.15% in an attention deficit hyperactivity disorder (ADHD) classification task using a multi-modality 3-D CNN model with TL between modalities. TL has also been applied in the early diagnosis of Alzheimer’s disease (AD) [30], [31]. In this case, although TL is not from a different dataset but rather from a subset of the same dataset, they obtained better accuracy in multi-site diagnostic prediction [32] with TL compared to without. Vakli et al. used resting-state functional connectivity (rs-FC) to predict age in healthy individuals and found that accuracy and generalizability of the model on their in-house dataset improved when they used a model that had been pre-trained using larger public databases such as Nathan Kline Institute-Rockland Sample (NKI-RS) dataset [33].

In contrast to these earlier studies, we are proposing to “transfer” the learning from healthy control data (obtained from HCP [34]) to situations where one has to predict whether a given subject is healthy or has a mental disorder (ASD in our case). The target domain (in our case, ABIDE data) is defined as the space to which the classifier pre-trained in the source domain (in our case, HCP data) is applied for classification. For the target domain, we utilized data from subjects with ASD and healthy controls available in the ABIDE dataset [35], pre-processed with a pipeline identical to that used for HCP data. The main idea is that by leveraging a large dataset of healthy controls, we can learn neural representations of mental health and transfer this knowledge to distinguish individuals with mental disorders from controls more effectively.

C. Data Oversampling Using Generative Models

Even though TL can address overfitting and enhance the generalizability of model, the class imbalance issue caused by TL cannot be ignored. Class imbalance arises when a dataset has large samples in one major class, and relatively smaller samples in other classes. Different from computer vision, natural language processing or other fields, publicly available large neuroimaging databases containing healthy control participants are available as compared to databases containing subjects with mental disorders. For example, ABIDE I and II [15] ($N \approx 2000$) are available for ASD and the Alzheimer’s Disease Neuroimaging Initiative (ADNI) [30] ($N \approx 3000$) for Alzheimer’s. In contrast, several datasets are available for healthy controls including HCP [34] ($N \approx 1200$), Healthy Brain Network (HBN) [36] ($N \approx 2500$) and U.K. Biobank [37] ($N > 10000$). Pooling these will provide us with over 10,000 healthy subjects, which is an order of magnitude more than the subjects available for any mental disorder. Therefore, any transfer learning from healthy datasets (as explained earlier) would create a scenario of unbalanced datasets during classification [38], wherein the DL model is trained by many more healthy control class samples as compared to patient class samples. This biases the model towards features from the major class, thereby impacting the overall error rate [39], [40], [41]. This situation is aggravated in TL and needs to be addressed.

To balance the dataset, the commonly used resampling techniques include undersampling the majority class data [42] or

oversampling the minority class data [43]. Generally, the undersampling works better than oversampling [44], but the latter does not lead to loss of data from the original dataset. The most widely used oversampling approach is synthesizing new examples and is called Synthetic Minority Oversampling Technique (SMOTE) developed by Chawla et al. [43]. The major steps of SMOTE include selecting samples closed in the latent feature space, drawing a separate line between the samples and then synthesizing new samples at a point along the line. However, SMOTE is known to be less effective on high-dimensional data such as gene expression microarray data [45]. Therefore, it is unlikely to work well on high dimensional fMRI data. Different from the traditional SMOTE method, with the development of generative modeling, Variational Auto-encoder (VAE) [46] and Generative Adversarial Networks (GAN) [47] for balancing classes have been successfully applied across various fields [48], [49], [50], [51]. The deep learning variant of SMOTE, known as DeepSMOTE [52], employs an encoder-decoder framework coupled with SMOTE as an oversampling technique, and has demonstrated its ability to generate high-quality images. Nonetheless, the utilization of DeepSMOTE with functional MRI data remains unclear. The details of VAE will be introduced below. In short, a generative model can learn the true underlying distribution of input observations in latent space during training, and can subsequently be applied to generate synthetic observations that closely resemble those of the training dataset [53]. By comparing different approaches, Fajardo et al. [53] found that a VAE based model outperformed GAN on benchmark MNIST (Modified National Institute of Standards and Technology database) image dataset. Wang et al. [54] developed a novel generative VAE model to generate synthetic data as a data augmentation strategy to achieve state-of-the-art 90.4% accuracy on high resolution ImageCLEF-DA images, and also reduce the divergence of input data distributions. Therefore, in this study, we utilized a VAE model to generate synthetic FC features from minority ASD class, and then use the synthetic data to balance the large-scale healthy control samples transferred from the HCP database.

D. CNN-Based VAE Model

Unlike conventional auto-encoders, VAE is a generative model that can train a generative representation from input data and generate new samples such as text, image, or language from the latent sample distribution [25], [26], [46]. Learning a generative representation from input data could potentially aid in improving generalizability of a CNN, something that we aim to achieve using TL. One group used VAE as an encoding model to analyze visual cortex fMRI data [55]. They found that the VAE model decodes video reconstruction activity in a more convenient way by transforming fMRI activity into low dimensional VAE latent space representations. Since VAE learns to model the data, researchers can sample from the distribution and generate new input data samples. Different from above mentioned studies, the goal of pre-training the model in the proposed TL approach is not just learning a compressed version of fMRI data, but also to learn the probability density across different data distributions because we have fMRI data from

different domains (or sites). Thus, the application of VAE in TL as applied to fMRI data is among our novel contributions.

For ASD classification, we applied a CNN model driven by FC features as discussed in Zou et al. [29]. The data-driven CNN has a lower dimension and less parameters than normal CNN, and hence is considered robust against overfitting [56], [57]. CNN has also been used in Alzheimer's classification, and has been proven to be an advanced supervised learning model useful in classifying Alzheimer's from controls in the ADNI data [30]. To implement a complete TL model, we transferred the encoder from VAE to CNN and replaced the decoder with fully connected layers for classification. Similar to prior fMRI research, the FC adjacency matrix was treated as a normal square image input to the designed convolutional filters. The kernels utilized in the encoder-decoder architecture of our model were inspired by Kawahara et al.'s model [56], known as Brain-NetCNN. These kernels were specifically designed to capture the interrelationships among brain regions and to detect patterns within connectivity matrices. Although previous studies have used TL to develop state-of-the-art learning models in different contexts [27], [28], most of these studies have not validated the effect of the TL model they applied on classification. To validate the process of how TL can improve the prediction, we tested the relationship between HCP sample size in the pre-training model and classification accuracy, hypothesizing that including more healthy subjects from HCP will improve classification accuracy in ABIDE data.

E. Stacked Auto-Encoder (SAE) in Model Pre-Training

In many previous DL-based studies, various kinds of hierarchical offline pre-training models were widely used [58], [59], [60]. Considering the complicated properties of fMRI neuroimaging data, a well-trained unsupervised learning model can extract essential features from the higher dimensional structure of neuroimaging data [58]. Prior research showed that features extracted by the stacked auto-encoder (SAE) are robust and efficient [58]. Another research report indicated that a pre-trained denoising AE can partially address the heterogeneity among data collected from different sites in ABIDE [61], [62]. Among the abovementioned studies, most have built a sparsity model against overfitting in high dimensional neuroimaging datasets [58], [63]. The sparsity of AE is critical in enhancing the generalizability of the features extracted. However, the drawback of sparsity is that it introduces a bias and reduces the power of neural networks. Thus, sparsity is an intrinsic trade-off between bias and overfitting of prediction models. The most commonly applied methods to increase the sparsity of CNN models are L1-regularization, L2-regularization, and drop-off layers [64]. Many studies have applied sparse deep neural networks to analyze fMRI data, thereby controlling the sparsity in various ways [57], [63], [65], [66], [67]. For example, measures of ANOVA in one schizophrenia study revealed that the effects of sparsity control in pre-trained hierarchical layers to the error rates were statistically significant (Bonferroni-corrected $p < 10^{-7}$; d.f. = 999) [60]. Therefore, to enhance the features extraction ability and address overfitting, we trained sparse CNN layers with L1-regularization [68] in a stacked AE

framework in this study. among data collected from different sites in ABIDE [61], [62].

F. LRP in Feature Identification and Interpretation

The interpretation of prediction results obtained from DL is a critical challenge in most studies attempting neuroimaging-based diagnostics. Unlike statistical and traditional machine learning models such as the support vector machine, the mechanism by which the DL model classifies and predicts the result remains a closed box [69], [70]. To better explain the basis of the results, previous studies using rs-FC for ASD classification [71], [72] have used the LRP (Layer-Wise Relevance Propagation) algorithm to identify the most discriminative FC features in ASD classification. The major advantage of the LRP method is that it can generate a pixel-wise scaled heat map (in the FC matrix space) to construct the contribution of each FC value towards the final classification result. In the end, on the basis of the contribution map of FC features, we can infer which neural mechanisms (based on FC) may be most predictive of ASD diagnostic status.

Various approaches may be used to interpret the results obtained by artificial neural networks, i.e., identify features that are discriminable across groups and likely led to the accuracy obtained. In some neuroimaging studies, the discriminative features are identified from the weights of shallow layers, while more abstract features are in the deeper layers of DNN [73]. Therefore, we applied LRP to run through all layers from the deepest output layer to the shallowest one. This type of feature interpretation approach is absent in TL neuroimaging studies so far. However, the LRP technique has been used in neuroimaging in other contexts. For example, a study used bi-directional LSTM to model the spatial dependencies of brain activity within and across brain slices [74]. The LRP algorithm was utilized based on the LSTM decoder to decompose the states decoding decision, and it accurately identified the physiologically appropriate associations between cognitive states and brain activity. LRP has also been applied to explain the subject-level classification outcome in structural MRI-based AD classification [73]. Multiple individual heat maps are constructed to determine their neurobiological relevance. Based on the well-developed LRP algorithm in the computer vision area, we designed a compact permutation statistical analysis method to identify biomarkers relevant for ASD classification.

Overall, we present a TL framework that addresses unmet needs in the field of neuroimaging based diagnostic classification as described above. Our main contributions can be summarized as follows:

- 1) We designed a stacked autoencoder model with L1-regularization to learn the compressed representations of healthy controls in the source domain (HCP + ABIDE datasets) in an unsupervised manner. Then the pretrained model was fine-tuned on the remaining ABIDE datasets to perform classification task.
- 2) To mitigate the challenge of class imbalance prevalent in transfer learning, particularly where the number of healthy

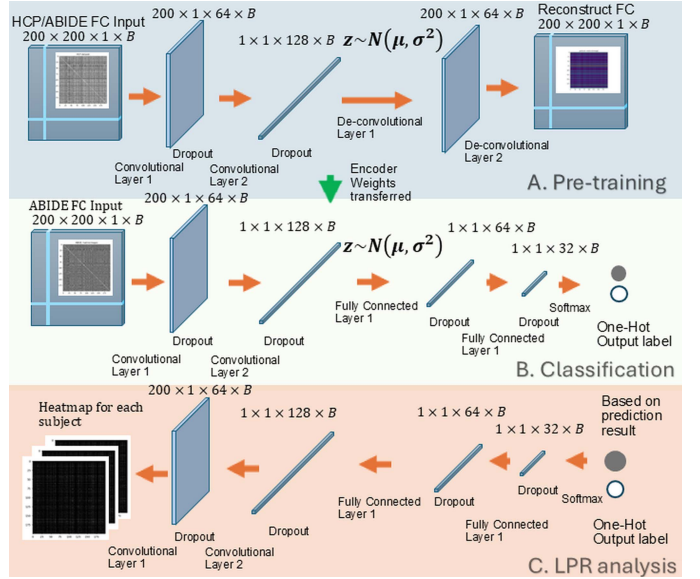


Fig. 1. The architecture of the VAE-CNN transfer learning framework, with different types of layers such as dropout layer, 1-D convolutional layer and de-convolve layer. There are three major components of this framework. (A) An example of a batch of HCP FC matrices input into the VAE model (B refers to the batch size of subjects). The offline VAE model consists of two stacked convolutional layers as an encoder, and two de-convolve layers as the decoder. (B) Then the weights of the pre-trained encoder are transferred into the FC-adaptive CNN for ASD classification. The deconvolved and reconstructed layers in the VAE model are replaced by the fully-connected classifier in the CNN model applied to ABIDE for diagnostic prediction. (C) After classifier training, we applied LRP analysis backward from identified ASD output to the input layer. The FC heat maps of each subject are constructed for FC biomarker identification in ASD prediction.

controls significantly outnumbers the patients with disorders, we introduced a generative VAE framework. This framework learns the representations of ASD samples and synthesizes ASD labels to construct a balanced dataset.

- 3) We introduce an interpretable method called the LRP algorithm, which is able to identify key connectivity patterns relevant to predicting the disease.

In the following section, we provide a detailed description of the methods employed.

II. MATERIALS AND METHODS

1) *Overview:* We illustrate the proposed VAE-CNN framework in Fig. 1. It consists of three major components: a pre-trained model to learn the encoder from unlabeled training data, a supervised learning model to predict labels from the ASD dataset, and an interpretation method to identify features in the data that have high discriminability between ASD and controls. As shown in Fig. 1, during model training in the source domain, we used the VAE on ASD and controls not only using ABIDE training data, but also controls from HCP data. We applied a CNN model in the target domain (ABIDE) for classification between ASD and controls, with transferred parameters initializing the pre-trained VAE model from the source domain. Classification performance was evaluated in terms of change in classification accuracy with an increasing number of

healthy subjects from HCP used during pre-training. The inputs were FC matrices from each subject. LRP was used to provide mechanistic and semantic insights underlying the classification decision.

A. Data Acquisition and Preprocessing

We used rs-fMRI data from the publicly available ABIDE I dataset [35] (18 sites). We employed binary classification between controls and ASD, i.e., Asperger's subjects in the dataset belonged to ASD despite having separate diagnosis. We did so because the number of subjects with Asperger's was quite small ($N = 93$) compared to ASD ($N = 339$) and controls ($N=556$) and we did not want to use a three-way classification problem with unbalanced datasets since TL was the focus of the study. Demographic information of participants is summarized in Supplemental Table.S1, including the distribution of subjects across individual sites and genders. All data were pre-processed using a standard pre-processing pipeline in the DPARSF toolbox [75]. This included the removal of the first five volumes, slice timing correction, motion correction and co-registration to the standard MNI space. Nuisance signals such as low-frequency drifts, motion parameters, and white matter and cerebrospinal fluid signals were regressed out. Time series were temporally filtered by a 0.01–0.1 Hz band-pass filter.

ABIDE dataset was our target domain since we were primarily interested in predicting whether a given subject has ASD or not. We split the 895 ABIDE subjects (selected after discarding a few subjects not meeting inclusion criteria) into training and testing datasets in the following ways: (i) site matched split: subjects were drawn randomly so that training ($N = 705$) and testing datasets ($N = 190$) were matched on non-imaging measures, and (ii) site mismatched split: training data from 11 sites and testing data from 7 remaining sites. This split was done to test the model's robustness to different sources of training and testing datasets, and hence potentially different distributions in training and testing data. In multi-site neuroimaging datasets such as ABIDE, subject-wise cross-validation schemes that randomly split participants without explicitly accounting for acquisition site or demographic structure can yield optimistic performance estimates due to shared site-related variance; in this sense, cross-validation may overestimate performance relative to evaluations using independent, site-mismatched test data that better reflect real-world clinical deployment [76]. Therefore, we preferred independent test data to evaluate the model.

For the source domain, we utilized the pre-processed HCP rs-fMRI data obtained directly from their website ($N=1097$, includes up to the S1200 release) [34], [77], [78]. ABIDE and HCP datasets are distinct on several aspects, including subject distribution, age/gender difference, pathology and pre-processing. This distinction is also a source of diversity among healthy controls that the classifier would be exposed to via TL, improving its ability to generalize to the larger population.

Since the dimensionality of fMRI data is very high, we extracted mean time series from 200 regions of interest (ROIs) as defined by the popular Craddock-200 atlas [79]. These 200 functionally homogenous ROIs cover the entire brain. BOLD fMRI is a convolution of latent (unmeasured) neural activity

and the hemodynamic response function (HRF). The HRF, which measures neurovascular coupling, varies between brain regions and individuals [80], [81]. This variability confounds FC estimates in healthy individuals [82], [83] as well as in subjects.

B. VAE-CNN Model

VAE is an unsupervised learning autoencoder model that has recently gained popularity. Typically, VAE is applied as a latent variable generative model to learn latent feature representation from unlabeled data, and then generate new data from it [46]. Here, we pre-trained VAE as a feature extraction model without supervision. Similar to conventional AE, VAE has an encoder and a decoder. The input FC data are referred to as x and latent feature representation is referred to as z . The probabilistic encoder is $q_\phi(z|x)$, which is parameterized as ϕ . Given a data point x , it produces a Gaussian distribution over the possible values of the latent variable z from which the data point x could have been generated. In a similar way, we will refer to $p_\theta(z|x)$ as a probabilistic decoder; given a latent variable z , it produces a distribution over the possible corresponding values of x . The marginal likelihood of a data point and the loss of evidence lower bound (ELBO) $\mathcal{L}^{\text{ELBO}}$ are constructed as follows:

$$\log p_\theta(x) = \text{D}_{\text{KL}}(q_\phi(z|x) \parallel p_\theta(z|x)) + \mathcal{L}^{\text{ELBO}}(\phi, \theta; x) \quad (1)$$

We need to maximize the ELBO loss, which is similar to minimizing the minus ELBO loss. Thus, the loss function of the VAE deep network can be written as:

$$\mathcal{J}^{(n)} = -\mathbb{E}_{z \sim q_\phi(z|x)}[\log p_\theta(x|z)] + \text{D}_{\text{KL}}(q_\phi(z|x) \parallel p(z)) \quad (2)$$

where the first item is a reconstructed loss. Similar to traditional deterministic AEs, the expectation of the log-likelihood is that the input image can be generated based on the sampled values of z from the inferred distribution $q_\phi(z|x)$. When $q_\phi(z|x)$ is a multivariate normal distribution with unknown expectations μ and variances σ^2 , the objective function is differentiable with respect to $(\theta, \phi, \mu, \sigma)$. The second term is the Kullback-Leibler (KL) divergence between approximate posterior distribution $q_\phi(z|x)$ that the encoder network maps the original data space into, as well as the pre-specified prior. In case of continuous latent variables, prior is typically assumed as Gaussian $\mathcal{N}(0, 1)$.

We propose that a VAE can outperform a normal AE for our context. However, VAE is difficult to optimize because the sampling in latent representation is not differentiable. Once we have the VAE network architecture defined, we re-parameterize it to make the VAE model trainable. Re-parameterization can stretch the encoded standard deviation with additional random noise, which propagates the reconstruction error to the encoder. Therefore, the tuning of parameters of VAE were optimized by using an Adam optimizer [84].

We constructed the VAE encoder with two stacked convolutional layers. Each convolutional layer is especially designed for FC input data. The details of this design are elaborated later. In addition, the decoder is constructed by two stacked deconvolution layers [85], [86]. Ours is the first to study apply VAE for extracting representative features from HCP FC data [87], [88] in the source domain and then use the pre-trained model in the target domain classification task using ABIDE.

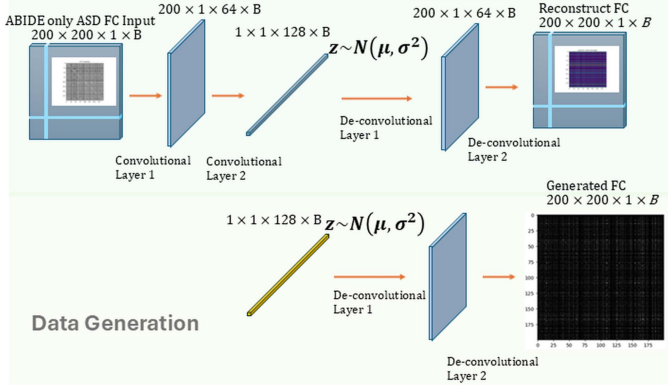


Fig. 2. Procedure for Synthetic Data Generation Using a VAE Trained on ABIDE ASD Data. This approach was adopted to solve the class imbalance between controls and ASD arising from transfer learning of control samples from the HCP data. No dropout layer was applied in the model.

C. VAE Pre-Training and Data Generation Model

As mentioned before, we use the VAE for pre-training and data oversampling. The idea is to let the VAE model learn latent representations of the data in an unsupervised way. In doing so, it is imperative to provide training examples from both classes. We used training data identified from ABIDE. As mentioned before, this is a total of 705 subjects, consisting of 277 ASD subjects and 428 controls subjects. However, just using the ABIDE training data in pre-training and then proceeding to classification by a CNN model will be the traditional approach devoid of any transfer learning. In order to bring in the component of transfer learning, we considered additional healthy control training examples from HCP database, consisting of 1097 control subjects. Therefore, in total, we had 1525 (1097 from HCP plus 428 from ABIDE) control subjects and 277 (from ABIDE) ASD subjects in training. This leads to class imbalance of 1248 samples (1525-277). In order to alleviate this class imbalance, we let the VAE learn latent representations of ASD data and generate an additional 1248 synthetic ASD FC matrices. The procedure, illustrated in Fig. 2, solves the problem of class imbalance while simultaneously enabling transfer learning from the HCP data. This approach could in-principle be used to add any number of healthy control samples that are more widely available publicly and hence is a key component and innovation in our work.

D. CNN Classification Model

Our FC-adaptive CNN classifier (i.e. a CNN model that has been adapted to suit FC features) consists of two 1D convolutional layers with the first layer for row connectivity convolution and the second layer for column connectivity convolution [56], [89], [90]. Each convolutional layer uses the ReLU activation function [91], and the output layer uses the Softmax function to obtain one-hot final result (a vector of each prediction with the predicted class as one and others as zeros). In addition, we applied dropout layers after all convolutional and fully connected layers to increase the sparsity of the deep neural network [92], [93].

The architecture of the FC-adaptive CNN is introduced in Fig. 1. To apply regularization, it includes multiple well-documented layers. Similar to the E2N layer in the BrainNetCNN model [56], the kernel size of the first-row convolutional layer is 200×1 . The kernel size of the second column convolutional layer is 1×200 . No pooling layer is included in FC-adaptive CNN. The first one has a weight size of (128,64) and bias size of (64, 1). The second layer has a weight size of (64, 32) and bias size of (32, 1). The third layer has a weights size of (32, 2) and bias size of (2, 1). Drop-out layers are applied after each convolutional layer. Each layer randomly shut down 0.7 proportion of neurons in training iterations. The Softmax layer can calculate the probability array into a one-hot labels array. The predicted labels were then compared with the ground truth labels to measure the classification accuracy. The final cost includes the training loss and L1 regularization term.

E. Transfer Learning Performance Estimation

To measure the performance of our TL framework, we test the ASD classifier under four approaches (Fig. 3). (1) As a baseline method (Fig. 3(A)), we followed an auto-encoder based DL approach similar to prior research [94], [95], [96], with only the ABIDE dataset as source domain data being used in training the VAE pre-train model. The training dataset included both healthy control (light blue, $N=428$) and ASD subjects (light Orange, $N = 277$) from the ABIDE dataset. Then we used the CNN classifier to test the model on ABIDE test dataset ($N = 190$). (2) To more specifically investigate the effect of TL (Fig. 3(B)), we used the combined HCP and ABIDE dataset in VAE training ($N = 1525$) for each of ASD and control groups, with the ASD group augmented by the VAE generative model as explained before. For each batch of HCP and ABIDE, we input HCP data first, which includes only healthy controls (dark blue), followed by the ABIDE dataset. (3) To measure the effect of source domain dataset size on TL-based prediction (Fig. 3(C)), we input only half of the source domain data ($N=548$) instead of 1097) and corresponding synthetic data in the pre-train model. (4) To test the capability of the TL model to potentially compensate for smaller sample size of the target domain data (Fig. 3(D)), we input only half of the training samples in ABIDE ($N = 350$ instead of 705). The estimation method was applied to both site-matched and site-mismatched datasets to measure the robustness of the model to multi-site effect.

Classifier performance was estimated by three diagnostic metrics: classification accuracy, balanced accuracy, and area under receiver operating characteristic curve (AUC). Classification accuracy is measured as % labels correctly predicted by the classifier on unseen test data. Given that ASD and control classes are unbalanced within both training and testing datasets, we also calculated balanced accuracy measured by normalizing true positive and true negative predictions by the number of positive and negative samples (TPR and TNR, respectively) and then calculating the mean of both values [97]:

$$\text{Balanced Accuracy} = (\text{TPR} + \text{TNR})/2 \quad (3)$$

In addition, ROC (receiver operating characteristic) reflects the diagnostic ability of a binary classifier system when its cutoff

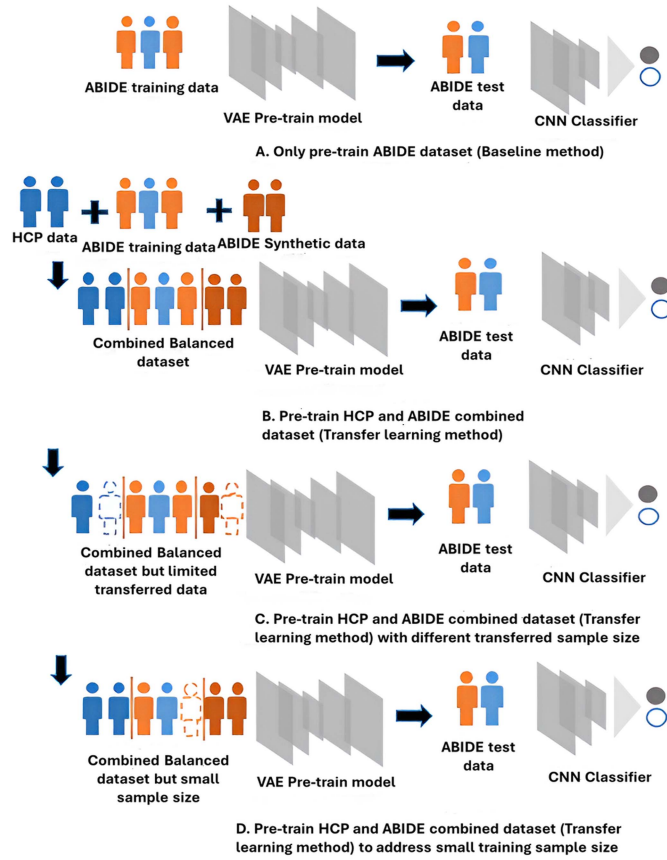


Fig. 3. Illustration of two strategies for ASD classification: (A) For the baseline method, we input ABIDE training and test datasets to the VAE-CNN framework. (B) For the transfer learning approach, we input the HCP and ABIDE balanced datasets into the VAE pre-train model and tested the classifier on the ABIDE test dataset. (C) To investigate the effect of transferred data size, we used half of combined balanced dataset of the previous approach and repeated the procedure. (D) To investigate the effect of small sample size, we used half the original training data in the transfer learning approach and repeated the procedure. Color code: light blue for ABIDE healthy subjects, light orange for ABIDE ASD subjects, dark blue for HCP healthy subjects and dark orange for ABIDE synthetic ASD subjects.

varies. AUC (area under ROC curve) was implemented using DL libraries Tensorflow [98] and PyTorch [99].

F. Layer-Wise Relevance Propagation (LRP) Algorithm

After training the deep network, we applied the LRP algorithm to investigate input features important for classification [100]. The core idea of LRP is to trace back the contribution to the final output neuron layer by layer. The basic principle of LRP is that the total relevance of an output node to a class is conserved by each layer. Each of the nodes in the l^{th} layer that contribute to the activation of a node j in the next $(l+1)^{th}$ layer a part of the relevance in node j : $R_{(l+1)}^j$. In other words, the relevance of a node in a layer is the sum of all relevance the nodes in previous layer contributed to:

$$\sum_i R_{l,l+1}^{i \rightarrow j} = R_{l+1}^j \quad (4)$$

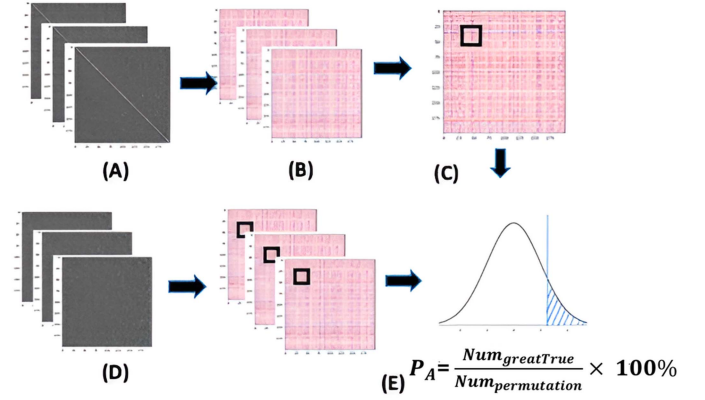


Fig. 4. Identifying FC features with high predictive ability using the LRP algorithm. (A) FC matrices from test subjects were input to the VAE-CNN transfer learning model. (B) The FC heat map for each subject was constructed using the LRP algorithm. (C) The true relevance score was calculated as the mean relevance across all subjects within a group. (D) Using 1000 randomizations, surrogate FC maps were created by random shuffling, and their heat maps were created using the same procedure used on the test dataset. (E) The final p-value for each FC feature was calculated from the true values and their corresponding surrogate distributions.

There are different ways in which the relevance can be distributed over the input neuron i , and different rules for how to distribute the relevance have been proposed [101]. In this paper, for better visualization, we apply the ω^2 rule [100] to define the propagation of relevance from the $(l+1)^{th}$ layer to the l^{th} layer. The ω^2 rule formula is given by [102]:

$$R_i = \sum_j \frac{\omega_{ij}^2}{\sum_i \omega_{ij}^2} R_j \quad (5)$$

Here, ω_{ij} is the vector of weight parameters that connects neuron x_i with x_j . R_i and R_j represent the relevance scores of neurons x_i and x_j . The propagation rule consists of redistributing relevance according to the square magnitude of the weights and pooling relevance across all neurons j . This rule is also valid for $R_j = 0$, where the actual point x_i is already a root, and for which no relevance needs to be propagated. For further details, please refer to similar implementations in earlier work [103] and [70]. A PyTorch implementation of the LRP algorithm was developed for the current work and is available on GitHub (<https://github.com/moboehle/Pytorch-LRP>).

G. Feature Identification

Input features were a 200×200 FC adjacency matrix, and output was a one-hot prediction array. We calculated the relevance values that propagate backward from output of one label to input features layer by layer. Then, because we are interested in FC features with best predictive ability, the output of ASD prediction was traced back to the input layer by LRP.

The procedure to identify FC biomarkers is illustrated in Fig. 4. The final heat map obtained from the LRP algorithm consisted of relevance scores of each feature in the input for a given subject. The mean relevance values of each feature across subjects (within a given group) were calculated as a heat

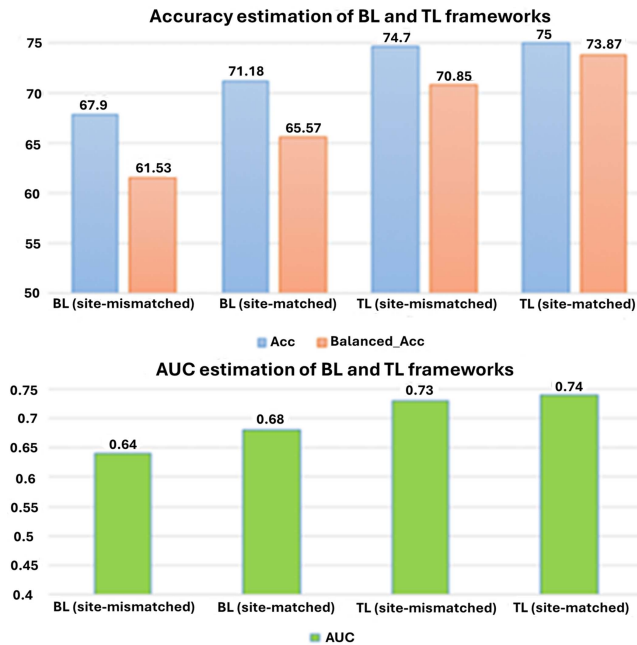


Fig. 5. Results obtained from ABIDE test data with and without transfer learning for both site-matched and site-mismatched cases (left: classification accuracies and right: total area underneath ROC curve (AUC)). Training and testing data were from different sites in the site-mismatched case, and drawn proportionally in the site-matched case. We found higher accuracy, balanced accuracy and AUC with transfer learning (TL) compared to baseline (BL) for all cases.

map representing the group. Then we used a permutation test to identify the statistical significance of the relevance values. We generated 1000 surrogate FC maps by randomly selecting functional connectivity data from the held-out test ABIDE dataset and shuffling subject labels within the test set only, while keeping the trained model fixed. These surrogate data also went through the exact same pipeline as the original test data and the LRP algorithm gave surrogate heat maps in each case. Thus, for each feature, we obtained a null distribution of relevance values that was not influenced by any training data, feature selection, or model fitting. By comparing the true relevance value with this surrogate distribution, we obtained a p-value for each input feature. The p-value of feature A is calculated as:

$$P_A = \frac{Num_{greatTrue}}{Num_{permutation}} \times 100\% \quad (6)$$

where $Num_{greatTrue}$ is the number of permutation values greater than true value, $Num_{permutation}$ is the permutation number (=1000). The significant features were determined in the heat maps based on an FDR-corrected p-value threshold of 0.05. To further check whether the identified features had stronger or weaker connectivity in ASD as compared to controls, we used two sample t-tests. These were then visualized using the BrainNet Viewer toolbox [104] in MATLAB.

III. RESULTS AND DISCUSSION

Fig. 5 compares the classification performance between baseline classification and the proposed TL framework. To address

the multi-site issue, we compared accuracies between site-matched and site-mismatched test datasets. In the site-matched case, the training and test data had subjects drawn proportionately from all sites. In the site-mismatched case, the training data were drawn from a few sites and the test data were from entirely different sites. In both cases, we found that the TL approach achieved higher accuracy (by about 7%) versus baseline classification. Accuracy with the site-mismatched dataset was also lower than with the site-matched dataset.

The AUC in Fig. 5 shows that transfer learning using the combined VAE-CNN classification model can achieve 0.73 AUC on site-mismatched dataset and 0.74 AUC on site-matched dataset, outperforming those obtained without TL. While classification with site-mismatched data deteriorated performance in the traditional baseline model in comparison to site-matched data (as expected [19]), TL helped the site-mismatched data catch up with the site-matched case by inclusion of more HCP data in the TL model. Thus, even though HCP data are from single site, it helps the model learn the variability in the healthy control population from a source other than ABIDE. This improves the generalizability (and hence, the performance) of the classifier. The performance improvement on unseen test data could in-principle be further boosted by exposing the model to more variability in both or either classes.

While it is encouraging to see TL's superior performance, a critical question is whether it is dependent on the sample size in the HCP source domain. In order to answer this question, we repeated the entire analysis pipeline using only half of the original HCP samples. We found (Fig. 6 and Fig. 7) improved performance with increased sample size of the source domain HCP data for both site-matched and site-mismatched cases.

This is encouraging because it indicates that there is further scope to improve performance in the future by adding more healthy control data in the source domain using other public databases that are freely available. To investigate the effect of the size of the target domain dataset on performance, we used only half of the ABIDE samples ($N = 350$) as training data. The bar chart in Fig. 8 illustrates that AUC from the smaller target domain training sample size is lesser than that from the full training sample size ($N = 705$), because small training sample sizes cannot fully represent the overall data distribution in the test dataset. However, with TL in the VAE-CNN model, the difference in balanced classification accuracy between small and normal datasets was reduced from around 6% to 3%. The same result is also reflected in the AUC plot.

Finally, the LRP algorithm identified FC features that were important for classification. Among connections identified as having significant relevance ($p < 0.05$, FDR corrected), those with hyper-connectivity in ASD as compared to controls are shown in Fig. 9 and those with hypo-connectivity are shown in Fig. 10. Overall, there were more ASD connections with hypo-connectivity compared to hyper-connectivity. This is not surprising given that hypo-connectivity has been an established hallmark of ASD [105], [106], [107], [108]. Significant hyper-connectivity was mainly observed in local pathways within the frontal lobe (Fig. 9), while hypo-connectivity was observed in a larger number of inter-hemisphere and inter-lobular connections

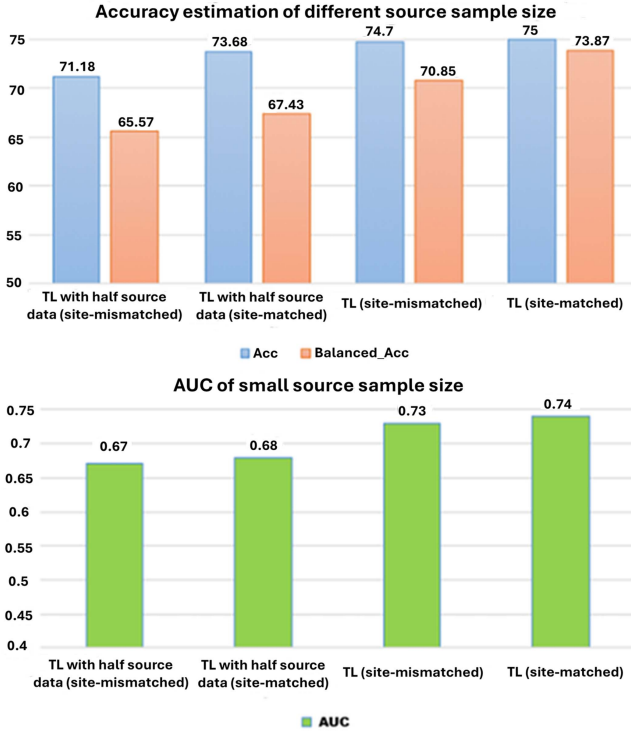


Fig. 6. Results obtained by using only half the samples in HCP source domain and its comparison with those obtained by using the full HCP sample in the source domain (left: classification accuracies and right: total area underneath ROC curve (AUC)). The classification accuracy is shown in blue, the balanced accuracy as red and AUC as green.

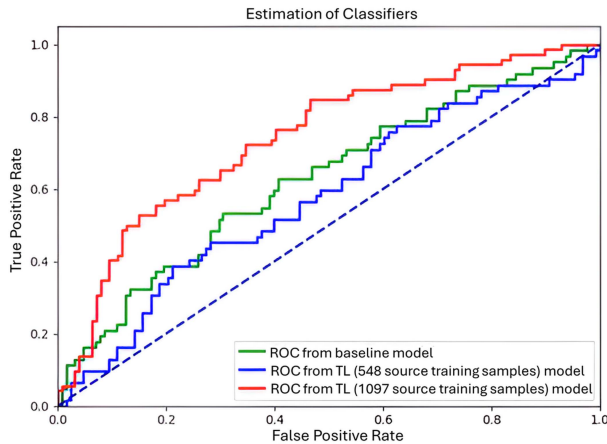


Fig. 7. ROCs obtained from site-mismatched test data with and without TL approaches. In the TL approach, the ROC curve achieved better shape when using the full HCP sample as source domain compared to using only half of the HCP samples as the source domain.

in ASD (Fig. 10). This is in agreement with the hypothesis [108] that ASD is characterized by local hyper-connectivity and distant hypo-connectivity. In Fig. 10, many paths connect superior frontal regions as well as temporal gyrus and lingual gyrus. This anterior-posterior FC disruption is a commonly observed pattern in rs-fMRI studies of ASD [109]. The medial temporal lobe network, which incorporates the insula, and shows reduced

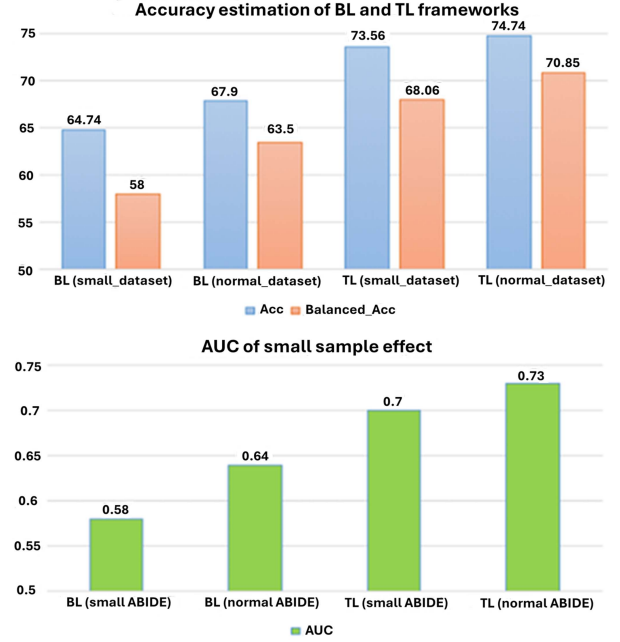


Fig. 8. Results obtained by using only half the samples in ABIDE target domain and its comparison with those obtained by using the full ABIDE sample in the target domain (left: classification accuracies and right: total area underneath ROC curve (AUC)). The classification accuracy is shown in blue, the balanced accuracy as red and AUC as green.

TABLE I
COMPARISON OF VARIOUS MACHINE LEARNING MODELS WITH OUR PROPOSED MODEL ON THE ABIDE-I DATASET

Study	Model	Accuracy (%)
Abraham et al. [111], 2017	SVC	68.00%
Almuqhim et al. [112], 2022	SAENet	70.80%
Cui et al. [113], 2023	Dual Graph - GCN	72.00%
Dong et al. [114], 2024	Multiview-Transformer	72.08%
Yan et al. [115], 2021	GAN	73.74%
Deng et al. [116], 2022	3D-CNN	74.53%
Our approach	TL + VAE + CNN	75.00%

inter-network connectivity, has also been found through ICA analysis of FC data [110].

To illustrate how our proposed model compares with previously applied machine learning methods, Table I provides an overview of other machine learning models that have been applied to the whole ABIDE I dataset. Compared to the traditional method (SVC [111], 68%), our proposed model achieves a substantially higher accuracy, improving performance by 7%. While Almuqhim et al. [112] also employed a sparse autoencoder (SAE), the integration of transfer learning in our framework led to a further gain of approximately 3% (75% vs. 72%). Our model also outperforms more recent deep learning approaches, including dual graph GCN (Cui et al. [113], 72%), multiview Transformer (Dong et al. [114], 72.08%), GAN (Yan et al. [115], 73.74%), and 3D-CNN (Deng et al. [116], 74.53%). These results highlight that the combination of transfer learning, VAE, and CNN provides a robust improvement over both classical and modern machine learning methods when applied to the entire ABIDE-I dataset.

with the type of machine learning models proposed here is an interesting topic for future research.

Although the knowledge transfer from the HCP to the ABIDE dataset has shown promising success in enhancing classification accuracy, it is necessary to validate the model's effectiveness across other datasets, such as ADHD-200 [120]. This additional testing will prove the model's effectiveness across diverse imaging sites. Furthermore, our findings demonstrate that increasing the number of healthy controls from the other dataset enhance the impact of transfer learning. Incorporating the U.K. Biobank [37], a dataset comprising over 10,000 subjects, into our model may further strengthen our hypothesis. We have yet to evaluate our proposed model with recent advancements in deep learning, such as Generative Adversarial Networks (GANs), which have demonstrated effectiveness in domain adaptation for classification tasks [121]. They should be included in the future works to truly assess the effectiveness of our model.

In a classification task, as done in our manuscript, we are predicting discrete variables such as diagnostic status. However, behavioral variables are continuous and require a regression model (and not a classification model) to be predicted. While using the proposed framework in a regression model is feasible, it is beyond the scope of the current manuscript.

V. CONCLUSION

In summary, for both site matched and mismatched splits, TL with VAE-CNN classifier outperforms traditional CNN without TL. Notably, the accuracies we have obtained on the independent test data are higher than those obtained from cross-validation in ABIDE [32], [71] (even though cross-validation over-estimates performance [19]), as well as on independent test data using traditional machine learning [19] and other TL approaches [32]. TL seems to alleviate deterioration of performance due to site-mismatched split, which may help in multi-site studies. Widespread hypo-connectivity in inter-hemispheric and inter-lobar connections with sporadic (local) hyper-connectivity in the right hemisphere (especially frontal lobe) in ASD are in line with literature [109].

The results show that the TL method outperforms ASD classification in test data as compared to other state-of-the-art methods, especially with small training sample size. The convolutional VAE model is robust against data divergence across different sites and between different data sources. We demonstrated the applicability of TL within a deep learning framework for utilizing larger samples of available healthy control data to improve generalizability and accuracy of diagnostic classification in ASD, as well as reduce the harmful effects of inter-site variability on classification. In addition, we identified brain FC patterns, which corroborated with previously published FC impairments in ASD.

REFERENCES

- [1] D. H. Geschwind and P. Levitt, "Autism spectrum disorders: Developmental disconnection syndromes," *Curr. Opin. Neurobiol.*, vol. 17, no. 1, pp. 103–111, 2007.
- [2] R. M. Jones and C. Lord, "Diagnosing autism in neurobiological research studies," *Behav. Brain Res.*, vol. 251, pp. 113–124, 2013.
- [3] G. S. Dichter, "Functional magnetic resonance imaging of autism spectrum disorders," *Dialogues Clin. Neurosci.*, vol. 14, no. 3, pp. 319–351, 2012.
- [4] M. R. Arbabshirani, S. Plis, J. Sui, and V. D. Calhoun, "Single subject prediction of brain disorders in neuroimaging: Promises and pitfalls," *Neuroimage*, vol. 145, pp. 137–165, 2017.
- [5] Y. Bengio et al., "Learning deep architectures for AI," *Found. Trends Mach. Learn.*, vol. 2, no. 1, pp. 1–127, 2009.
- [6] V. Shukla and S. Choudhary, "Deep learning in neural networks: An overview," in *Deep Learning in Visual Computing and Signal Processing*. New York, NY, USA: Apple Academic Press, 2022, pp. 29–53.
- [7] M. H. Al-Adhaileh, "Diagnosis and classification of Alzheimer's disease by using a convolution neural network algorithm," *Soft Comput.*, vol. 26, no. 16, pp. 7751–7762, 2022.
- [8] X. Li et al., "BrainGNN: Interpretable brain graph neural network for fMRI analysis," *Med. Image Anal.*, vol. 74, 2021, Art. no. 102233.
- [9] N. Qiang et al., "Functional brain network identification and fMRI augmentation using a VAE-GAN framework," *Comput. Biol. Med.*, vol. 165, 2023, Art. no. 107395.
- [10] J. Hu, Y. Huang, and S. Dong, "Transformer and snowball graph convolution learning for brain functional network analysis," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, 2023, pp. 2809–2816.
- [11] C. Jing, C. Gong, Z. Chen, B. Lei, and S. Wang, "TA-GAN: Transformer-driven addition-perception generative adversarial network," *Neural Comput. Appl.*, vol. 35, no. 13, pp. 9579–9591, 2023.
- [12] J. Donahue et al., "Long-term recurrent convolutional networks for visual recognition and description," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 2625–2634.
- [13] D. Bahdanau, J. Chorowski, D. Serdyuk, P. Brakel, and Y. Bengio, "End-to-end attention-based large vocabulary speech recognition," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, 2016, pp. 4945–4949.
- [14] A. Graves, A.-r. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, 2013, pp. 6645–6649.
- [15] A. Di Martino et al., "Enhancing studies of the connectome in autism using the Autism brain imaging data exchange II," *Sci. Data*, vol. 4, no. 1, pp. 1–15, 2017.
- [16] D. Durstewitz, G. Koppe, and A. Meyer-Lindenberg, "Deep neural networks in psychiatry," *Mol. Psychiatry*, vol. 24, no. 11, pp. 1583–1598, 2019.
- [17] D. Wen, Z. Wei, Y. Zhou, G. Li, X. Zhang, and W. Han, "Deep learning methods to process fMRI data and their application in the diagnosis of cognitive impairment: A brief overview and our opinion," *Front. Neuroinform.*, vol. 12, 2018, Art. no. 23.
- [18] S. Vieira, W. H. Pinaya, and A. Mechelli, "Using deep learning to investigate the neuroimaging correlates of psychiatric and neurological disorders: Methods and applications," *Neurosci. Biobehavioral Rev.*, vol. 74, pp. 58–75, 2017.
- [19] P. Lanka, D. Rangaprakash, M. N. Dretsch, J. S. Katz, T. S. Denney, and G. Deshpande, "Supervised machine learning for diagnostic classification from large-scale neuroimaging datasets," *Brain Imag. Behav.*, vol. 14, pp. 2378–2416, 2020.
- [20] B. Mwangi, T. S. Tian, and J. C. Soares, "A review of feature reduction techniques in neuroimaging," *Neuroinformatics*, vol. 12, pp. 229–244, 2014.
- [21] J. A. Nielsen et al., "Multisite functional connectivity MRI classification of autism: Abide results," *Front. Hum. Neurosci.*, vol. 7, 2013, Art. no. 599.
- [22] J. A. Turner et al., "A multi-site resting state fMRI study on the amplitude of low frequency fluctuations in Schizophrenia," *Front. Neurosci.*, vol. 7, 2013, Art. no. 137.
- [23] A. Di Martino et al., "The autism brain imaging data exchange: Towards a large-scale evaluation of the intrinsic brain architecture in autism," *Mol. Psychiatry*, vol. 19, no. 6, pp. 659–667, 2014.
- [24] X. Yang, M. S. Islam, and A. A. Khaled, "Functional connectivity magnetic resonance imaging classification of autism spectrum disorder using the multisite abide dataset," in *Proc. IEEE EMBS Int. Conf. Biomed. Health Inform.*, 2019, pp. 1–4.
- [25] K. Sarinaporn and M. Kubat, "Combining subclassifiers in text categorization: A DST-based solution and a case study," *IEEE Trans. Knowl. Data Eng.*, vol. 19, no. 12, pp. 1638–1651, Dec. 2007.
- [26] J. Zhang, W. Li, P. Ogunbona, and D. Xu, "Recent advances in transfer learning for cross-dataset visual recognition: A problem-oriented perspective," *ACM Comput. Surv.*, vol. 52, no. 1, pp. 1–38, 2019.

- [27] C. Doersch, "Tutorial on variational autoencoders," 2016, *arXiv:1606.05908*.
- [28] S. Semeniuta, A. Severyn, and E. Barth, "A hybrid convolutional variational autoencoder for text generation," *Empirical Methods Natural Lang. Process.*, Copenhagen, Denmark, 2017.
- [29] L. Zou, J. Zheng, C. Miao, M. J. McKeown, and Z. J. Wang, "3D CNN based automatic diagnosis of attention deficit hyperactivity disorder using functional and structural MRI," *IEEE Access*, vol. 5, pp. 23626–23636, 2017.
- [30] E. Hosseini-Asl, G. Gimel'farb, and A. El-Baz, "Alzheimer's disease diagnostics by a deeply supervised adaptable 3D convolutional network," *Front. Biosci.*, vol. 23, no. 3, pp. 584–596, 2018.
- [31] B. Cheng, M. Liu, D. Zhang, D. Shen, and A. D. N. Initiative, "Robust multi-label transfer feature learning for early diagnosis of Alzheimer's disease," *Brain Imag. Behav.*, vol. 13, pp. 138–153, 2019.
- [32] H. Li, N. A. Parikh, and L. He, "A novel transfer learning approach to enhance deep neural network classification of brain functional connectomes," *Front. Neurosci.*, vol. 12, 2018, Art. no. 491.
- [33] P. Vakli, R. J. Deák-Meszlényi, P. Hermann, and Z. Vidnyánszky, "Transfer learning improves resting-state functional connectivity pattern analysis using convolutional neural networks," *Gigascience*, vol. 7, no. 12, 2018, Art. no. giy130.
- [34] S. M. Smith et al., "Resting-state fMRI in the human connectome project," *Neuroimage*, vol. 80, pp. 144–168, 2013.
- [35] C. Craddock et al., "The neuro bureau preprocessing initiative: Open sharing of preprocessed neuroimaging data and derivatives," *Front. Neuroinform.*, vol. 7, no. 27, 2013, Art. no. 5.
- [36] L. M. Alexander et al., "An open resource for transdiagnostic research in pediatric mental health and learning disorders," *Sci. Data*, vol. 4, no. 1, pp. 1–26, 2017.
- [37] K. L. Miller et al., "Multimodal population brain imaging in the UK biobank prospective epidemiological study," *Nature Neurosci.*, vol. 19, no. 11, pp. 1523–1536, 2016.
- [38] M. McFarquhar, "Testable hypotheses for unbalanced neuroimaging data," *Front. Neurosci.*, vol. 10, 2016, Art. no. 270.
- [39] A. Estabrooks, T. Jo, and N. Japkowicz, "A multiple resampling method for learning from imbalanced data sets," *Comput. Intell.*, vol. 20, no. 1, pp. 18–36, 2004.
- [40] L. Yuan et al., "Multi-source feature learning for joint analysis of incomplete multiple heterogeneous neuroimaging data," *NeuroImage*, vol. 61, no. 3, pp. 622–632, 2012.
- [41] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.
- [42] G. Haixiang, L. Yijing, J. Shang, G. Mingyun, H. Yuanyue, and G. Bing, "Learning from class-imbalanced data: Review of methods and applications," *Expert Syst. Appl.*, vol. 73, pp. 220–239, 2017.
- [43] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [44] R. Blagus and L. Lusa, "Smote for high-dimensional class-imbalanced data," *BMC Bioinf.*, vol. 14, pp. 1–16, 2013.
- [45] L. Lusa et al., "Evaluation of smote for high-dimensional class-imbalanced microarray data," in *Proc. 11th Int. Conf. Mach. Learn. Appl.*, 2012, vol. 2, pp. 89–94.
- [46] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.
- [47] I. Goodfellow et al., "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, vol. 27, pp. 2672–2680.
- [48] M. Lopez-Martin, B. Carro, A. Sanchez-Esguevillas, and J. Lloret, "Conditional variational autoencoder for prediction and feature recovery applied to intrusion detection in IoT," *Sensors*, vol. 17, no. 9, 2017, Art. no. 1967.
- [49] G. Mariani, F. Scheidegger, R. Istrate, C. Bekas, and C. Malossi, "BAGAN: Data augmentation with balancing GAN," 2018, *arXiv:1803.09655*.
- [50] G. Douzas and F. Bacao, "Effective data generation for imbalanced learning using conditional generative adversarial networks," *Expert Syst. Appl.*, vol. 91, pp. 464–471, 2018.
- [51] P. Shamsolmoali, M. Zareapoor, L. Shen, A. H. Sadka, and J. Yang, "Imbalanced data learning by minority class augmentation using capsule adversarial networks," *Neurocomputing*, vol. 459, pp. 481–493, 2021.
- [52] D. Dablain, B. Krawczyk, and N. V. Chawla, "DeepSMOTE: Fusing Deep Learning and Smote for Imbalanced Data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 9, pp. 6390–6404, Sep. 2023.
- [53] V. A. Fajardo et al., "On oversampling imbalanced data with deep conditional generative models," *Expert Syst. Appl.*, vol. 169, 2021, Art. no. 114463.
- [54] Q. Wang, F. Meng, and T. P. Breckon, "Data augmentation with norm-VAE for unsupervised domain adaptation," 2020, *arXiv:2012.00848*.
- [55] K. Han et al., "Variational autoencoder: An unsupervised model for encoding and decoding fMRI activity in visual cortex," *NeuroImage*, vol. 198, pp. 125–136, 2019.
- [56] J. Kawahara et al., "BrainNetCNN: Convolutional neural networks for brain networks; towards predicting neurodevelopment," *NeuroImage*, vol. 146, pp. 1038–1049, 2017.
- [57] R. J. Meszlényi, K. Buza, and Z. Vidnyánszky, "Resting state fMRI functional connectivity-based classification using a convolutional neural network architecture," *Front. Neuroinform.*, vol. 11, 2017, Art. no. 61.
- [58] X. Han, Y. Zhong, L. He, P. S. Yu, and L. Zhang, "The unsupervised hierarchical convolutional sparse auto-encoder for neuroimaging data classification," in *Proc. 8th Int. Conf. Brain Inf. Health*, London, U.K., 2015, pp. 156–166.
- [59] H.-I. Suk, S.-W. Lee, D. Shen, and A. D. N. Initiative, "Latent feature representation with stacked auto-encoder for AD/MCI diagnosis," *Brain Struct. Function*, vol. 220, pp. 841–859, 2015.
- [60] F. Li, L. Tran, K.-H. Thung, S. Ji, D. Shen, and J. Li, "A robust deep model for improved classification of AD/MCI patients," *IEEE J. Biomed. Health Inform.*, vol. 19, no. 5, pp. 1610–1616, Sep. 2015.
- [61] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn.*, 2008, pp. 1096–1103.
- [62] A. S. Heinsfeld, A. R. Franco, R. C. Craddock, A. Buchweitz, and F. Meneguzzi, "Identification of autism spectrum disorder using deep learning and the abide dataset," *NeuroImage: Clin.*, vol. 17, pp. 16–23, 2018.
- [63] J. Kim, V. D. Calhoun, E. Shim, and J.-H. Lee, "Deep neural network with weight sparsity control and pre-training extracts hierarchical features and enhances classification performance: Evidence from whole-brain resting-state functional connectivity patterns of schizophrenia," *NeuroImage*, vol. 124, pp. 127–146, 2016.
- [64] A. Payan and G. Montana, "Predicting Alzheimer's disease: A neuroimaging study with 3D convolutional neural networks," 2015, *arXiv:1502.02506*.
- [65] M. Khosla, K. Jamison, A. Kucyeski, and M. R. Sabuncu, "Ensemble learning with 3D convolutional neural networks for functional connectome-based prediction," *NeuroImage*, vol. 199, pp. 651–662, 2019.
- [66] Y. Li, J. Liu, J. Huang, Z. Li, and P. Liang, "Learning brain connectivity sub-networks by group-constrained sparse inverse covariance estimation for alzheimer's disease classification," *Front. Neuroinform.*, vol. 12, 2018, Art. no. 58.
- [67] X. Chen, J. He, R. Lawrence, and J. G. Carbonell, "Adaptive multi-task sparse learning with an application to fMRI study," in *Proc. SIAM Int. Conf. Data Mining*, 2012, pp. 212–223.
- [68] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. Roy. Stat. Soc. Ser. B: Stat. Methodol.*, vol. 58, no. 1, pp. 267–288, 1996.
- [69] M. Alber et al., "iNNvestigate neural networks!" *J. Mach. Learn. Res.*, vol. 20, no. 93, pp. 1–8, 2019.
- [70] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," 2013, *arXiv:1312.6034*.
- [71] Z. Sherkatghanad et al., "Automated detection of autism spectrum disorder using a convolutional neural network," *Front. Neurosci.*, vol. 13, 2020, Art. no. 1325.
- [72] Y. Bai, Z. Pascal, W. Hu, V. D. Calhoun, and Y.-P. Wang, "Biomarker identification through integrating fMRI and epigenetics," *IEEE Trans. Biomed. Eng.*, vol. 67, no. 4, pp. 1186–1196, Apr. 2020.
- [73] L. Kohoutová et al., "Toward a unified framework for interpreting machine-learning models in neuroimaging," *Nature Protoc.*, vol. 15, no. 4, pp. 1399–1435, 2020.
- [74] A. W. Thomas, H. R. Heekeren, K.-R. Müller, and W. Samek, "Analyzing neuroimaging data through recurrent deep learning models," *Front. Neurosci.*, vol. 13, 2019, Art. no. 1321.

- [75] C. Yan and Y. Zang, "DPARSF: A MATLAB toolbox for" pipeline" data analysis of resting-state fMRI," *Front. Syst. Neurosci.*, vol. 4, 2010, Art. no. 1377.
- [76] L. He, H. Li, S. K. Holland, W. Yuan, M. Altabe, and N. A. Parikh, "Early prediction of cognitive deficits in very preterm infants using functional connectome data in an artificial neural network framework," *NeuroImage: Clin.*, vol. 18, pp. 290–297, 2018.
- [77] K. Ugurbil et al., "Pushing spatial and temporal resolution for functional and diffusion MRI in the human connectome project," *Neuroimage*, vol. 80, pp. 80–104, 2013.
- [78] S. M. Smith et al., "Advances in functional and structural MR image analysis and implementation as FSL," *Neuroimage*, vol. 23, pp. S208–S219, 2004.
- [79] R. C. Craddock, G. A. James, P. E. Holtzheimer III, X. P. Hu, and H. S. Mayberg, "A whole brain fMRI atlas generated via spatially constrained spectral clustering," *Hum. Brain Mapping*, vol. 33, no. 8, pp. 1914–1928, 2012.
- [80] D. A. Handwerker, J. M. Ollinger, and M. D'Esposito, "Variation of bold hemodynamic responses across subjects and brain regions and their effects on statistical analyses," *Neuroimage*, vol. 21, no. 4, pp. 1639–1651, 2004.
- [81] G. K. Aguirre, E. Zarahn, and M. D'Esposito, "The variability of human, bold hemodynamic responses," *Neuroimage*, vol. 8, no. 4, pp. 360–369, 1998.
- [82] D. Rangaprakash, G.-R. Wu, D. Marinazzo, X. Hu, and G. Deshpande, "Hemodynamic response function (HRF) variability confounds resting-state fMRI functional connectivity," *Magn. Reson. Med.*, vol. 80, no. 4, pp. 1697–1713, 2018.
- [83] D. Rangaprakash, G.-R. Wu, D. Marinazzo, X. Hu, and G. Deshpande, "Parameterized hemodynamic response function data of healthy individuals obtained from resting-state functional MRI in a 7T MRI scanner," *Data Brief*, vol. 17, pp. 1175–1179, 2018.
- [84] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [85] W. Shi et al., "Is the deconvolution layer the same as a convolutional layer?," 2016, *arXiv:1609.07009*.
- [86] M. D. Zeiler, D. Krishnan, G. W. Taylor, and R. Fergus, "Deconvolutional networks," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, 2010, pp. 2528–2535.
- [87] D. P. Kingma, S. Mohamed, D. Jimenez Rezende, and M. Welling, "Semi-supervised learning with deep generative models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, vol. 27, pp. 3581–3589.
- [88] S. R. Bowman, L. Vilnis, O. Vinyals, A. M. Dai, R. Jozefowicz, and S. Bengio, "Generating sentences from a continuous space," 2015, *arXiv:1511.06349*.
- [89] J. Ji and Y. Yao, "A novel CNN framework to extract multi-level modular features for the classification of brain networks," *Appl. Intell.*, vol. 52, pp. 6835–6852, 2022.
- [90] J.-H. Kim, Y. Zhang, K. Han, Z. Wen, M. Choi, and Z. Liu, "Representation learning of resting state fMRI with variational autoencoder," *NeuroImage*, vol. 241, 2021, Art. no. 118423.
- [91] A. F. Agarap, "Deep learning using rectified linear units (ReLU)," 2018, *arXiv:1803.08375*.
- [92] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.
- [93] S. Wager, S. Wang, and P. S. Liang, "Dropout training as adaptive regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2013, vol. 26, pp. 351–359.
- [94] H. Sewani and R. Kashef, "An autoencoder-based deep learning classifier for efficient diagnosis of autism," *Children*, vol. 7, no. 10, 2020, Art. no. 182.
- [95] T. Eslami, V. Mirjalili, A. Fong, A. R. Laird, and F. Saeed, "Asd-diagnet: A hybrid learning approach for detection of autism spectrum disorder using fMRI data," *Front. Neuroinform.*, vol. 13, 2019, Art. no. 70.
- [96] W. Yin, S. Mostafa, and F.-X. Wu, "Diagnosis of autism spectrum disorder based on functional brain networks with deep learning," *J. Comput. Biol.*, vol. 28, no. 2, pp. 146–165, 2021.
- [97] K. H. Brodersen, C. S. Ong, K. E. Stephan, and J. M. Buhmann, "The balanced accuracy and its posterior distribution," in *Proc. 2010 20th Int. Conf. Pattern Recognit.*, 2010, pp. 3121–3124.
- [98] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous distributed systems," 2016, *arXiv:1603.04467*.
- [99] A. Paszke et al., "Pytorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, vol. 32, pp. 8024–8035.
- [100] A. Binder, S. Bach, G. Montavon, K.-R. Müller, and W. Samek, "Layer-wise relevance propagation for deep neural network architectures," in *Proc. Inf. Sci. Appl.*, 2016, pp. 913–922.
- [101] M. Böhle, F. Eitel, M. Weygandt, and K. Ritter, "Layer-wise relevance propagation for explaining deep neural network decisions in MRI-based Alzheimer's disease classification," *Front. Aging Neurosci.*, vol. 11, 2019, Art. no. 194.
- [102] G. Montavon, S. Lapuschkin, A. Binder, W. Samek, and K.-R. Müller, "Explaining nonlinear classification decisions with deep Taylor decomposition," *Pattern Recognit.*, vol. 65, pp. 211–222, 2017.
- [103] G. Montavon, W. Samek, and K.-R. Müller, "Methods for interpreting and understanding deep neural networks," *Digit. Signal Process.*, vol. 73, pp. 1–15, 2018.
- [104] M. Xia, J. Wang, and Y. He, "BrainNet viewer: A network visualization tool for human brain connectomics," *PLoS One*, vol. 8, no. 7, 2013, Art. no. e68910.
- [105] N. Chaitra, P. Vijaya, and G. Deshpande, "Diagnostic prediction of autism spectrum disorder using complex network measures in a machine learning framework," *Biomed. Signal Process. Control*, vol. 62, 2020, Art. no. 102099.
- [106] M. A. Syed, Z. Yang, X. P. Hu, and G. Deshpande, "Investigating brain connectomic alterations in autism using the reproducibility of independent components derived from resting state functional MRI data," *Front. Neurosci.*, vol. 11, 2017, Art. no. 459.
- [107] G. Deshpande, L. E. Libero, K.R. Sreenivasan, H. D. Deshpande, and R. K. Kana, "Identification of neural connectivity signatures of autism using machine learning," *Front. Hum. Neurosci.*, vol. 7, 2013, Art. no. 670.
- [108] R. K. Kana, L. Q. Uddin, T. Kenet, D. Chugani, and R.-A. Müller, "Brain connectivity in Autism," *Frontiers Hum. Neurosci.*, vol. 8, 2014, Art. no. 349.
- [109] V. L. Cherkassky, R. K. Kana, T. A. Keller, and M. A. Just, "Functional connectivity in a baseline resting-state network in autism," *Neuroreport*, vol. 17, no. 16, pp. 1687–1690, 2006.
- [110] E. A. von dem Hagen, R. S. Stoyanova, S. Baron-Cohen, and A. J. Calder, "Reduced functional connectivity within and between 'social' resting state networks in autism spectrum conditions," *Social Cogn. Affect. Neurosci.*, vol. 8, no. 6, pp. 694–701, 2013.
- [111] A. Abraham et al., "Deriving reproducible biomarkers from multi-site resting-state data: An autism-based example," *NeuroImage*, vol. 147, pp. 736–745, 2017.
- [112] F. Almuqhim and F. Saeed, "ASD-SAE Net: A sparse autoencoder, and deep-neural network model for detecting autism spectrum disorder (ASD) using fMRI data," *Front. Comput. Neurosci.*, vol. 15, 2021, Art. no. 654315.
- [113] W. Cui et al., "Dynamic multi-site graph convolutional network for autism spectrum disorder identification," *Comput. Biol. Med.*, vol. 157, 2023, Art. no. 106749.
- [114] Q. Dong, H. Cai, Z. Li, J. Liu, and B. Hu, "A multiview brain network transformer fusing individualized information for autism spectrum disorder diagnosis," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 8, pp. 4854–4865, Aug. 2024.
- [115] D. Yan et al., "Improving brain dysfunction prediction by GAN: A functional-connectivity generator approach," in *Proc. IEEE Int. Conf. Big Data*, 2021, pp. 1514–1522.
- [116] J. Deng, M. R. Hasan, M. Mahmud, M. M. Hasan, K. A. Ahmed, and M. Z. Hossain, "Diagnosing autism spectrum disorder using ensemble 3D-CNN: A preliminary study," in *Proc. IEEE Int. Conf. Image Process.*, 2022, pp. 3480–3484.
- [117] T. He et al., "Deep neural networks and kernel regression achieve comparable accuracies for functional connectivity prediction of behavior and demographics," *NeuroImage*, vol. 206, 2020, Art. no. 116276.
- [118] B. J. Casey et al., "The adolescent brain cognitive development (ABCD) study: Imaging acquisition across 21 sites," *Devlop. Cogn. Neurosci.*, vol. 32, pp. 43–54, 2018.
- [119] W. E. Johnson, C. Li, and A. Rabinovic, "Adjusting batch effects in microarray expression data using empirical Bayes methods," *Biostatistics*, vol. 8, no. 1, pp. 118–127, 2007.
- [120] A. Consortium, "The ADHD-200 consortium: A model to advance the translational potential of neuroimaging in clinical neuroscience," *Front. Syst. Neurosci.*, vol. 6, 2012, Art. no. 62.
- [121] C. Han et al., "MadGAN: Unsupervised medical anomaly detection GAN using multiple adjacent brain MRI slice reconstruction," *BMC Bioinf.*, vol. 22, pp. 1–20, 2021.



an affiliated faculty with the Department of Psychological Sciences. He has authored or coauthored more than 120 peer-reviewed journal papers in the field of signal processing, neuroimaging, brain connectivity and machine learning. His research is funded by the U.S. Department of Defense, National Science Foundation and National Institutes of Health. Dr. Deshpande was the recipient of the numerous awards including the President's Outstanding Collaborative Units Award.

Gopikrishna Deshpande (Senior Member, IEEE) received the MS degree in electrical communication engineering from the Indian Institute of Science Bangalore, India, in 2003, and the PhD degree in biomedical engineering from Georgia Institute of Technology, USA, in 2007. From 2007 to 2010, he was first a postdoc and then a research faculty with Biomedical Imaging Technology Center, Emory University, USA. He is currently a professor with the AUMRI Research Center, Department of Electrical and Computer Engineering, Auburn University, USA, where he is also



for brain disease classification, brain connectivity analysis and exploring laminar fMRI techniques. He was a presenter at the ISMRM2023, representing a topic that focuses on the functional organization of the Default Mode Network (DMN) using refined-scale fMRI.

Nguyen Huynh received the BS degree in physics from the University of Science, Ho Chi Minh City, Vietnam, in 2018, and the MS degree in information technology from Purdue University, Fort Wayne, Indiana, USA, in 2020. He is currently working toward the PhD degree in electrical and computer engineering with Auburn University, Auburn, Alabama, USA. He is also as graduate teaching assistant and graduate research assistant with Auburn MRI Research Center, Auburn, Alabama, USA. His research interest includes applying machine learning to neuroimaging



Bonian Lu received the BS degree in electrical engineering from Hunan University, China, in 2014, and the PhD degree in electrical engineering from Auburn University, USA, in 2022. He was a Data Scientist with the industry, first with FDB Technology Inc., and is currently with FocusKPI, Inc., Union City, CA, USA. His research focuses on the application of machine learning methods to neuroimaging data for medical diagnostics.



Northwestern University, Chicago, where he worked on modeling neurovascular coupling using fMRI. He is currently a senior research fellow with the Martinos Center for Biomedical Imaging, Harvard Medical School, researching brain and spinal cord function using functional MRI (fMRI). His research focuses on to adopt perspectives from engineering to answer difficult neuroscience questions using brain imaging.

D Rangaprakash received the Master of Science degree in electrical communication engineering from the Indian Institute of Science, India, and the PhD degree in electrical and computer engineering from Auburn University, Alabama, in 2016, focusing on identifying fMRI biomarkers of post-traumatic stress disorder in soldiers. He was a postdoctoral researcher with the University of California, Los Angeles, studying time-varying brain connectivity in anxiety disorders and predicting clinical outcomes using machine learning. He was a postdoctoral research fellow with